International Scientific Physics Olympiad

T1. Polycrystal diffusion

Answer sheet

Part A. Temperature dependence of the diffusion coefficient (2.6 points)

A1 **0.2**

$v =$

$D(T) \propto$

A2 **0.4**

$\Delta r_{\text{sq}} =$

A3 **2.0**

T	t	N	Δr_{sq}	D

$E_g =$

Accuracy:

Part B. Effective diffusion coefficient (2.4 points)

B1 **0.4**

$\alpha =$

B2 **2.0**

t	D_{gb}/D	δ	α

$\alpha =$

Part C. Type B kinetics (5 points)

C1 **0.4**

C2 **4.0**

t	D_{gb}/D	δ	$\ln \bar{c}$

$\xi =$ $\eta =$

$\zeta =$

C3 **0.6**

$C =$